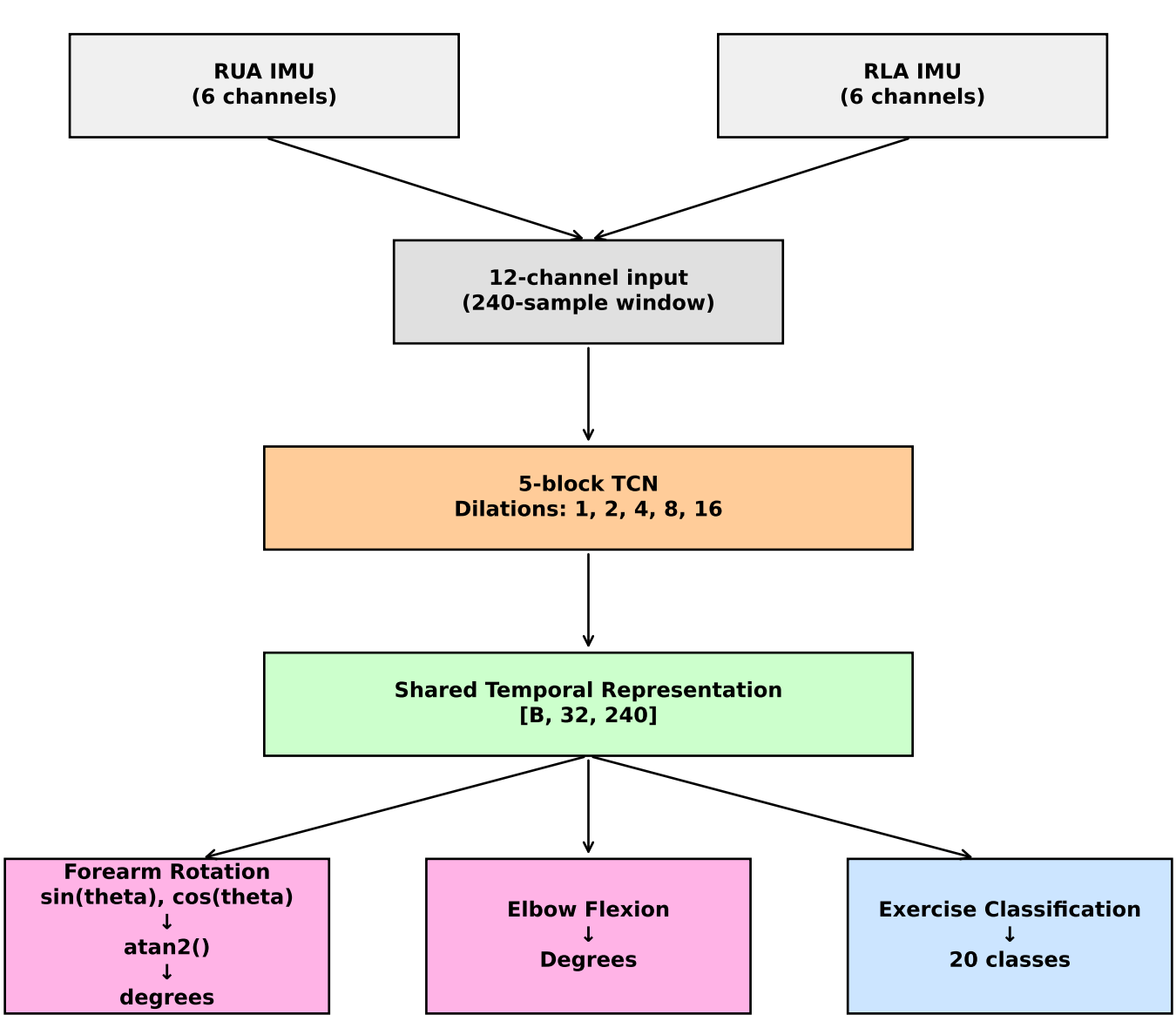
Multi-Task TCN Architecture
RUA IMU
(6 channels)
RLA IMU
(6 channels)
12-channel input
(240-sample window)
5-block TCN
Dilations: 1, 2, 4, 8, 16
Shared Temporal Representation
[B, 32, 240]
Forearm Rotation
sin(theta), cos(theta)
↓
atan2()
↓
degrees
Elbow Flexion
↓
Degrees
Exercise Classification
↓
20 classes